Qn. Set Code-7

Semester: 5th
Programme: B.Tech
Branch: CSE, CSCE, CSSE

AUTUMN END SEMESTER EXAMINATION-2024

5th Semester B.Tech

HIGH PERFORMANCE COMPUTING

CS30005 / CS 3010

(For 2023 (L.E), 2022 & Previous Admitted Batches)

Time: 2 Hours 30 Minutes | Full Marks: 50

Answer any FIVE questions.
Question paper consists of two SECTIONS i.e. A and B.
Section A is compulsory.
Attempt any Four question from Sections B.
The figures in the margin indicate full marks.
Candidates are required to give their answers in their own words as far as practicable and all parts of a question should be answered at one place only.

SECTION-A

1. Answer the following questions. [1 × 10]

(a) Write down the functions of following registers used in MIPS datapath ;

A, Imm, LMD, NPC

(b) A program runs in 18sec in computer A which has 2GHz clock. A computer designer builds a computer B that runs the same program in 11 sec. But there is a need of substantial increase of clock rate that causes computer B requiring 1.3 times as many clock cycles as computer A. What clock rate should we tell the designer to target?

(c) How register renaming is used to resolve WAR? Explain with example.

(d) From target and from fall through schemes, which is beneficial for branch taken scheme and also for branch untaken scheme? Justify your answer.

(e) There is 40 additions and each addition requires 4 clock cycles. An array and a vector processor have 4 Processing Elements (PE). The vector processor uses a 4 stage pipeline. Then what is the clock cycles required to finish in array and vector processor?

(f) What is cache coherence problem in a multi-processor? Give an example.

(g) Suppose that in 5000 memory references there are 165 misses in the first-level cache and 225 misses in the second-level cache. Calculate the global and local miss rates for both the caches.

(h) Calculate the percentage of miss prediction accuracy using 2-bit dynamic branch prediction scheme for the following outcome of a branch:

NT, T, NT, NT, NT, T, T, T, NT, NT, T, NT, T, NT, T, NT, T

Assume the initial prediction is strongly taken.

(i) Distinguish between super-scalar and super-pipeline processor.

(j) Differentiate between conflict miss and capacity miss with an example.

SECTION-B

2. (a) "Amdahl's Law Quantifies overall performance gain due to improve in a part of a computation." -: Justify & prove the statement for speed up overall. [5]

Assume that 25 % of your program is floating point operations and rest of the program is integer operations. A design team D1 proposes a speed up for the floating point operations by five times using a dedicated floating point hardware circuit. Another design team D2 proposes the speed up for integer operations by two times. Which design has the better performance?

(b) Briefly, Explain different MIPS instruction format (R type, I type, and J type) with examples from each type? [5]

3. (a) Distinguish between write through and write back. Assume a fully associative write back cache with many cache entries that starts empty. Below is a sequence of memory operations (the address is in square brackets). [5]

I. Readmem [500]

II. WriteMem [500]

III. ReadMem [400]

IV. WriteMem [300]

V. ReadMem [200]

VI. ReadMem [400]

VII. ReadMem [300]

VIII. WriteMem [300]

IX. WriteMem [200]

X. ReadMem [300]

XI. WriteMem [300]

What are the numbers of hits and misses when using no-write allocate versus write allocate scheme?

(b) Consider two processor A and B But both have ideal CPI of 1.5 clock cycle. Both have different stall cycles for data and control hazard which is given below. [5]

	Processor A	Processor B
Cycle time	150 nanoseconds	350 nanoseconds
Taken branch stalls	2	4
Not Taken branch stalls	2	4
Load use stalls	1	2
Store Stalls	1	2

program consists of 40% branches, 15% loads, 20% stores and 25% other instructions. Find out the CPI of both machines if 70% of branches are taken and 75% of loads are followed by a dependent instruction. Which machine is faster to execute the above program?

4. (a) Explain the various optimizations used for reducing the miss rate with suitable example. [5]

(b) Consider the information given in the table below. [5]

Functional Unit (FU)	# of FUs	EX cycles
Integer	**2**	**3**
Floating Point Multiply	**2**	**8**
Floating Point add	**2**	**2**
Floating point Divide	**1**	**50**

For the following MIPS code sequence show the instruction status, functional unit status and register result status using scoreboard approach

1. L.D F2, 8(R2)
2. L.D F6, 16(R3)
3. MUL.D F0, F2, F4
4. ADD.D F6, F8, F2
5. DIV.D F10, F0, F6

5. (a) Explain the working and implementation of snoopy based protocol with state diagram in details. [5]

(b) Explain loop unrolling with respect to the following example. [5]

```
for (i=90; i>0; i=i–1)
    a[i] = a[i] + K;
```

This code adds a scalar(K) to a vector. Here each element in the array is of type double, occupying 8 bytes. Assume the following latency in the example and branch delay of 1 clock cycle.

Instruction producing result	**Instruction using result**	***Latency in clock cycles***
FP ALU op	Another FP ALU op	3
FP ALU op	Store double	2
Load Double	FP ALU op	1
Load Double	Store double	0
Integer op	Integer op	0

if loop unrolling is performed for every three iterations. Determine how many clock cycles are required per iteration after scheduling the unrolled loop?

6. (a) Compare and analyze direct mapped and set associative cache with a suitable mapping example. [5]

   (b) Explain how reservation stations CDB used in the Tomasulo's approach are useful in resolving the WAW and WAR hazard with an example. [5]

7. (a) Assume two cache memory specifications. A split cache which consists of 32 KB instruction cache and 32 KB Data cache and 64 KB unified cache. The cache memory hit time is of 3 clock cycles, miss penalty time is of 90 clock cycles. [5]

   For the above cache memory specifications a Simulator showed the following conditions like :-

   1) 5 miss per 1000 instructions for data cache.

2) 40 miss per 1000 instructions for instruction cache.

3) 45 miss per 1000 instructions for the unified cache.

If there are 30% Load/Store instructions then calculate the miss rate and average memory access time (AMAT) for split and unified cache memory.

(b) Write short note on *any two*: [5]

(i) Vector processor Architecture

(ii) VLIW processor Architecture

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